PEST MANAGEMENT DECISION GUIDE: GREEN AND YELLOW LIST

Diamondback moth on cabbage

Pluttela xylostella Sefasefa in Tonga language, Nyaja in Bemba languages



Adult moths do not cause damage (Lyle Buss, University of Florida).



Cabbage damaged by Diamondback moth larvae (A.M. Varela, icipe.)



Larval feeding (Photo by A.M. Varela, icipe.)

Prevention

- Plant cabbage in March to June, or before September which are sub optimal periods for diamond back moth.
- Rotate with non-cabbages such as onions, legumes for at least 1.5 months.
- Slash left-over cabbage stems after harvest to avoid regrowth which act as a diamond back moth breeding area.
- Plant Marigold, *Tagetes minuta* (local name Mutanda zyello), or tomatoes or pepper around the cabbage field to repel moths.

Monitoring

- Scout for small light greenish, sometimes grey-greenish, thin larvae when cabbage is at four leaf stage.
- Continue inspecting weekly especially on the underside of the leaves.
- Check the cabbage for small irregular holes or ”shot holes” in the leafs when the larvae are young thus, 3 to 5 mm long larvae.
- You may consider direct control when you notice 2 to 3 young larvae per plant.

Direct Control

- When you notice diamondback moth larvae, use overhead irrigation to drop larvae from cabbages.
- Release tiny *Cotesia vestalis* wasps to parasitize the larvae. You can get them from Z.A.R.I, Entomology Section, Mrs. Tembo.
- Spray cabbage with neem tree leaf extract: Soak 1kg pounded neem tree leaves in 5 litres of water 1 day. Then, the soaked pounded leaves are taken out and squeezed to get the extract drain back into the bucket. The extracted solution is sieved with a sieve or mutton cloth to get a clear solution for use in the sprayer. The extract is sprayed on the leaves after mixing with water in a 2:1 ratio. Add some drops of liquid soap as a sticker to obtain better coverage of leafs.
- Alternatively spray chilli mixed with water and few chopped pieces of bar soap or drops of liquid soap.
- Crush larvae manually when numbers are few.

Direct Control / Restrictions

Direct Control	Restrictions
• Note that newly emerged tiny larvae are hidden from sprays, but young 3 to 5 mm larvae can be controlled. Spraying old about 1 cm larvae is often too late to prevent damage.	
• When using a pesticide (even a botanical home-brew), always wear protective clothing. Follow the instructions on the product label, such as dosage, timing of application, pre-harvest interval, max number of sprays, restricted re-entry interval. Do not empty into drains.	
• WHO class II pesticides might not be allowed in local IPM schemes.	
• Always consult recent list of registered pesticides (ZEMA).	
• Lambda Cyhalothrin –based products (TRIGER 5 EC; MOZICON; ATS-ICON 10CS; BIOPLANT 5%EC BOLLPACK; KANGFU; LAMBDA 5EC; and many others). Pyrethroid group of pesticides.	• WHO toxicity class II (moderately hazardous); Pre-harvest interval (p.h.i.) 3- 7days; restricted re-entry interval (r.e.i.) 3 days after spray; max 2 sprays per season.
• Cypermethrin –based products (CYPERM ETHRIN 20%; SENTINEL 15%EC; and others). Pyrethroid group of pesticides. Often applied at 20-30ml per 20 litres, but double-check with product labels.	• WHO toxicity class II (moderately hazardous); P.h.i. 7days; r.e.i. 1 day after spray. Spay max 2 x with 7 day interval (1 x only when flowering). High risk to bees, very toxic to aquatic organisms. Does not work well during hot weather.

Researching Soils, Crops and Water in Zambia

Zambia

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